1 Inputs and initialization

Specify model parameters

$T_{\rm eff}, \quad \log g, \quad \{n_i/n_{\rm ref}\}$

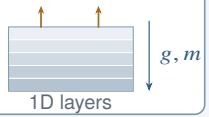
Effective temperature, gravity, composition

Construct the initial atmosphere

$P_{\rm g}(m) \simeq gm, \quad T(m)$

Gas pressure, temperature

m: column mass



2 Atmospheric iteration

Solve the equation of state and populations

$\rho(m), \quad n_e(m), \quad \{n_s(m)\}$

Mass density, electron density, species populations

Calculate opacity and optical depth

$\chi_\lambda = \kappa_\lambda + \sigma_\lambda, \qquad \tau_\lambda(m) = \int_0^m \chi_\lambda(m')\, dm'$

$\kappa_\lambda, \sigma_\lambda$: absorption, scattering; τ_λ: optical depth

Solve radiative and convective transport

$J_\lambda(m), \quad F_{\rm rad}(m), \quad F_{\rm conv}(m)$

Mean intensity, radiative flux, convective flux

Converged?

$F_{\rm rad}(m) + F_{\rm conv}(m) \simeq \sigma_{\rm SB} T_{\rm eff}^4$

Flux balance and a stable temperature structure

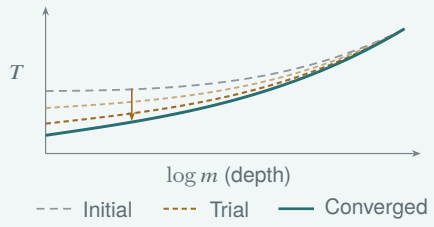
NO

Update the temperature

$T(m) \leftarrow T(m) + \delta T(m)$

Trust-region Newton step

Temperature relaxation



YES

3 Spectrum and model

Calculate the emergent spectrum

$F_\lambda = \int I_\lambda(\theta) \cos\theta \, d\Omega$

I_λ: specific intensity; F_λ: flux density

Fine wavelength grid

Return the model

$\{T(m), P_{\rm g}(m), \rho(m), n_e(m)\}, \; F_\lambda$

